In [25]:
```
import pandas as pd
dataset=pd.read_csv("Social_Network.csv")
dataset
dataset=pd.get_dummies(dataset,drop_first=True)
dataset=dataset.drop(columns=["User ID"])
dataset
x=dataset[["Age","EstimatedSalary","Gender_Male"]]
y=dataset[["Purchased"]]
print(y)
from sklearn.model_selection import train_test_split
x_train,x_test,y_train,y_test=train_test_split(x,y,test_size=1/6,random_state=42)
from sklearn.preprocessing import StandardScaler
sc=StandardScaler()
x_train=sc.fit_transform(x_train)
x_test=sc.transform(x_test)
```

```
     Purchased
0            0
1            0
2            0
3            0
4            0
..         ...
395          1
396          1
397          1
398          0
399          1

[400 rows x 1 columns]
```

In [26]:
```
dataset[["Purchased"]].value_counts()
dataset.shape
```

Out[26]:

```
(400, 4)
```

In [27]:
```
from sklearn.naive_bayes import GaussianNB
clf=GaussianNB()
clf.fit(x_train,y_train)
```

```
C:\Users\User\anaconda3\Lib\site-packages\sklearn\utils\validation.py:1365: DataConversionWarning:
A column-vector y was passed when a 1d array was expected. Please change the shape of y to
(n_samples, ), for example using ravel().
  y = column_or_1d(y, warn=True)
```

Out[27]:

▾ GaussianNB

▸Parameters

▸Fitted attributes

In [28]:
```
y_pred=clf.predict(x_test)
print(y_pred)
```

```
[1 1 0 1 0 0 1 0 0 0 0 1 0 0 0 0 1 1 0 1 0 0 1 1 0 1 0 0 1 0 0 0 1 0 1 0 0
 0 0 0 1 0 0 1 0 1 0 0 1 0 0 1 0 0 0 0 1 0 0 0 0 0 1 0 0 1 1]
```

In [30]:
```
from sklearn.metrics import confusion_matrix
r=confusion_matrix(y_pred,y_test)
print(r)
```

```
[[43  1]
 [ 2 21]]
```

In [31]:
```
from sklearn.metrics import classification_report
cr=classification_report(y_pred,y_test)
print(cr)
```

```
              precision    recall  f1-score   support

           0       0.96      0.98      0.97        44
           1       0.95      0.91      0.93        23

    accuracy                           0.96        67
   macro avg       0.96      0.95      0.95        67
weighted avg       0.96      0.96      0.95        67
```

In [ ]: